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Framework to completely bypass expensive DFT calculations via graph neural networks for vacancy formation energy predictions in FCC high entropy alloys



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The compositional complexity and chemical randomness of high entropy alloys (HEAs) make conventional atomic-scale calculations, such as density functional theory (DFT), prohibitively expensive for property prediction. One key property of interest is the vacancy formation energy (E_v^f), which plays a crucial role in diffusion and microstructure evolution. In this work, we present a machine learning (ML) framework that eliminates the need for DFT calculations by predicting E_v^f s in HEAs using models trained on binary and ternary alloys. Our approach first relaxes face-centered cubic (FCC) structures using a fine-tuned CHGNet model and then uses the resulting configurations as input into a crystal graph convolutional neural network (CGCNN) to predict both Bader charges and E_v^f s. Incorporating Bader charges as descriptors introduces DFT-informed electronic structure information into the model, significantly improving prediction accuracy compared to using elemental features alone. Furthermore, we demonstrate that the model's performance generalizes well to other alloy systems with minimal fine-tuning, offering a robust and efficient path toward high-throughput defect property prediction in complex alloys.

The electronic and atomic level chemical and structural complexities in high entropy alloys (HEAs) present a rich landscape of novel properties^{1–6}. These alloys are now at the forefront of alloy design. One of the characteristic features of HEAs is local lattice distortion (LLD) that arises due to the presence of multiple principal elements and their chemical randomness^{7–9}. LLD is considered to be responsible for enhanced alloy strength, as the moving dislocations are recursively inhibited by the ever-distorted energy landscape^{10,11}. In addition, the chemical randomness gives rise to a range of nearest-neighbor environments, causing each site to have a unique bonding environment. This results in large variations of atomic-level properties, such as point-defect energies, which can potentially enable improved diffusional-creep strength^{10,12,13}. Therefore, understanding atomic-level mechanisms in HEAs is critical to the design and development of alloys for future applications.

Over decades, density functional theory (DFT) has played a crucial role in developing such mechanistic understanding in dilute/conventional alloys. However, the enormous compositional space of HEAs poses

significant computational challenges for DFT. Data-science methods, such as machine learning (ML) models, are aptly beneficial. Recent ML studies have employed physics-based features to capture underlying structure-property relationships^{14–17}. Moreover, structure-based models, such as graph neural networks (GNNs), have been valuable because of their ability to include structural and atomic-level information in the model while also allowing for predictions on various system sizes^{18–22}. Some of these methods still rely on DFT relaxed structures as input to predict defect properties, which remains computationally expensive. In pursuit of further reducing computational costs, other studies have focused on machine learning interatomic potentials (MLIPs) such as ALIGNN²³, M3GNet²⁴ and CHGNet²². These models are trained on large DFT databases (e.g. Materials Project Database²⁵). However, these databases typically consist of ordered, defect-free structures. As such, the models lack relaxation information in the presence of point defects often leading to errors in predictions²⁰. Furthermore, because such models are trained on ordered structures, they lack the ability to accurately predict LLD present in HEAs.

In this work, we introduce a machine learning framework to overcome these limitations and entirely bypass DFT calculations to obtain accurate vacancy formation energies (E_v^f s). The framework is demonstrated in a Ni-Cu-Au-Pd face-centered cubic (FCC) alloys system. This is achieved by 1) reaching DFT level accurate relaxation and lattice distortion of HEAs through fine-tuning CHGNet, and 2) predicting DFT-based partial charges and including them as additional physics-based features in a modified crystal graph convolutional neural network (CGCNN¹⁹) based model to predict E_v^f s. Because this model framework uses the relaxed, defect-free supercell via CHGNet, the vacancy relaxation deficiency of MLIPs observed by Choudary et al.²⁰ is overcome, while still allowing DFT to be bypassed. We choose to use CHGNet as it is currently one of the top-performing models on MatBench and can easily be fine-tuned with user-generated data, perfect for application in HEAs. The models follow our previously developed PREDICT¹⁵ approach, in which the models are trained on simpler binary and ternary alloys to make predictions in complex HEAs.

Even though this work does not train the models on more than a handful of elements due to the limited availability of defect databases, we demonstrate that the model is capable of learning new information by fine-tuning on a small dataset with minimal data loss and greatly improved predictability. The new, smaller dataset introduced for fine-tuning towards Ni-Co-Cr is comprised of one of each of the following random alloys: Ni_{0.50}Co_{0.50}, Ni_{0.75}Cr_{0.25}, and Ni_{0.33}Co_{0.33}Cr_{0.33}. This framework and the models presented herein can be used as a baseline for further defect studies in HEAs to aid in alloy design while only requiring very few extra DFT calculations for large gains in predictability in other HEAs. These models are then used to validate charge predictions and defect energetics for other Ni-Co-Cr alloys from a short-range order structure database deposited in MPContribs by Walsh et al.²⁶

Results

Effects of Bader charge and atomic volume on vacancy formation energy

In FCC alloys after nearest-neighbor (NN) contributions, two key atomic-scale mechanisms govern the variability in E_v^f s: differences in atomic radii and differences in electronegativity²⁷. Variations in atomic size influence the local atomic volume around a vacancy, while electronegativity differences drive charge redistribution, captured through Bader charge (BC). These effects become especially significant in HEAs, where a broad distribution of local environments arises due to random mixing of elements.

Our previous work²⁷ demonstrated that E_v^f s are strongly dependent on the NN environment. Even within a fixed NN composition, that is, a vacancy surrounded by the same number and type of nearest neighbors, a range of E_v^f s can be observed²⁷. We found that for a given NN environment, vacancy sites with larger local atomic volumes exhibit higher E_v^f s. This trend is shown in Fig. 1a, c, e for random equiatomic Ni-Cu, Ag-Au, and Cu-Au alloys, respectively. In Ni-Cu (Fig. 1a) and Cu-Au (Fig. 1e), the E_v^f increases with volume, reflecting the significant atomic size mismatches between the constituent elements. However, this volume effect is absent in Ag-Au (Fig. 1c), where the atomic radii of Ag and Au are nearly identical, resulting in minimal variation.

The second mechanism—electronegativity difference—affects E_v^f s through charge transfer between the atoms, as captured by the Bader charge. Figure 1b, d, f shows E_v^f as a function of Bader charge for the same alloy systems. In Ni-Cu (Fig. 1b), where electronegativity differences are small, there is little to no correlation between Bader charge and formation energy. In contrast, trends appear in Ag-Au (Fig. 1d) and Cu-Au (Fig. 1f), where the more electronegative Au atoms gain charge and the less electronegative Ag or Cu atoms lose charge. These charge redistributions correspond to systematic variations in E_v^f s. Thus, both atomic volume and Bader charge serve as key descriptors for vacancy energetics in disordered alloys, although their relative influence is alloy dependent.

Because HEAs consist of several elements composed of a range of atomic sizes and electronegativities, it is expected that their E_v^f s ought to be significantly affected by the intertwined interplay of the two mechanisms.

Every lattice site essentially presents a unique set of NN features, which not only makes the quantitative but also the qualitative trends unpredictable. In addition, to trace the energy landscape, statistically sufficient sampling of the energy landscape requires a large number of formation energy DFT calculations that challenge the computational practicality of the approach. Thus, it is desirable to rely on machine learning to learn the complex behaviors in simpler binary and ternary alloys to predict them in the more complex quaternary or quinary HEAs. The framework of this machine learning approach is depicted in Fig. 2, where a relaxed CHGNet crystal structure is fed in as input to the Bader charge model. From here, both the structure and the predicted Bader charges are fed into the E_v^f model. Thus, DFT is bypassed via ML to predict E_v^f s.

Accurate lattice distortion predictions by fine-tuning CHGNet MLIP

In order to predict E_v^f s from ML, it is critical to first obtain accurate atomic relaxation. Specifically, in HEAs, it is important to capture the lattice distortion. This is an inherent feature of HEAs and arises due to the displacement of atoms from their ideal lattice sites¹¹. We initiate with the readily available MLIP CHGNet²² (v0.3.0) to relax the chemically random structures. The mean atomic displacement (MAD) is often used to quantify the distortion (see Methods, Eq. (5)). We validate the capability of the universal CHGNet v0.3.0 potential against the DFT results. The left vertical axis in Fig. 3a shows the MAD, whereas the right vertical axis shows atomic radii difference δ , which is an average difference in the atomic radii in multicomponent alloys, quantified by Zhang et al.²⁸. Figure 3a indicates that alloys with larger δ tend to have a higher MAD. The MAD data from DFT and CHGNet v0.3.0 is shown in circles and squares, respectively. A significant difference between the two is observed, which indicates that the pre-trained (off-the-shelf) CHGNet is not accurate against DFT for chemically random structures. That is understandable because the pre-trained model is mainly trained on ordered structures that do not consist of atomic displacements, as mentioned above.

Following our PREDICT approach, we fine-tune CHGNet with a database of binary and ternary random FCC alloys from a previous work²⁹ (see Methods) to accurately capture the MAD and lattice distortions of HEAs. This is achieved by freezing several model layers within CHGNet (see Methods). We also update the energies of our DFT database to match those of CHGNet, as shown in Supplementary Fig. S1. This is done so that the model minimizes the error on atomic forces and thus atomic positions (see Methods and Supplemental Information). The MAD from the fine-tuned CHGNet is shown by triangles in Fig. 3a, indicating a very good agreement with DFT. Note that the fine tuning is only done in binary and ternary alloys; this is enough to predict accurate MAD in the quaternary alloys.

Figure 3b shows the MAD with its standard deviation as error bars for the v0.3.0 pretrained CHGNet model, the fine-tuned CHGNet model, and DFT in quaternary alloys. The pretrained model has not only a larger MAD for each alloy but also a much larger standard deviation compared to the fine-tuned model and DFT. A similar analysis for binary and ternary alloys is presented in the Supplementary Fig. S4. Lastly, Fig. 3c shows the coefficient of determination (R^2) and the root mean square error (RMSE) for the MAD between the v0.3.0 pre-trained and the fine-tuned CHGNet models vs. DFT. The atomic relaxation is also evidenced in lattice parameters as shown in Supplementary Fig. S2.

Charge transfer prediction from modified crystal graph convolutional neural network (CGCNN)

The next step is to obtain the Bader charges of the atoms from the CHGNet relaxed structures. The electronegativity is a major driving force for charge transfer, and as such an element with a higher electronegativity will gain charge and vice-versa.

The electronegativities of the elements in this work are as follows: $\chi_{Au} > \chi_{Pd} > \chi_{Ni} \approx \chi_{Cu}$. Thus, for example, in a Ni-Cu-Au alloy, Au is expected to gain charge, while Ni and Cu are expected to lose charge, as shown in our previous study²⁷. Because charge transfer is determined primarily by

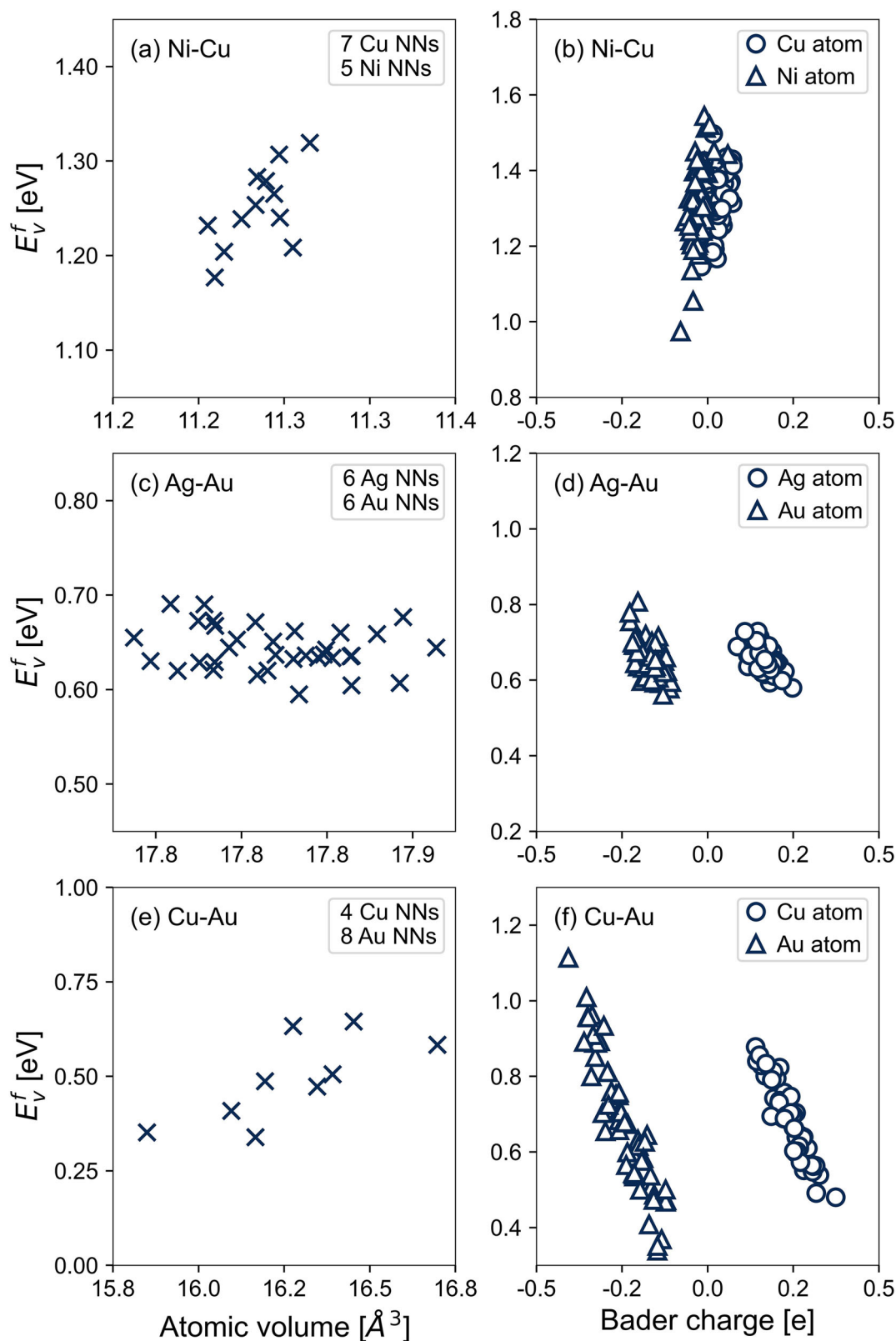
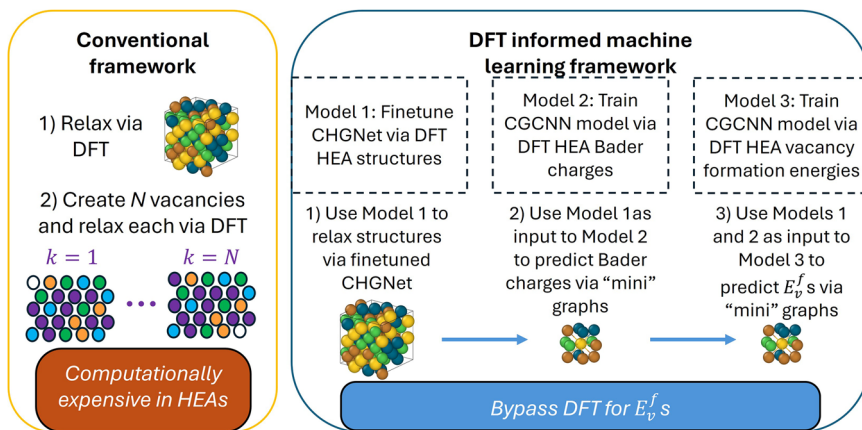


Fig. 1 | Factors that influence vacancy formation energies in alloys. **a, c, e** Atomic volume vs. E_v^f and **b, d, f** Bader charge vs. E_v^f in Ni-Cu, Ag-Au and Cu-Au alloys. In Ni-Cu and Cu-Au (**a** and **e**), the E_v^f increases with atomic volume due to significant differences in the atomic volumes of the two elements of the alloy, whereas in Ag-Au (in **c**), there is no correlation due to similar atomic volumes of

Ag and Au. Similarly, the electronegativity differences also affect E_v^f . In Ag-Au and Cu-Au (in **d** and **f**, respectively), E_v^f is correlated to Bader charge due to significant electronegativity differences between the two elements of the alloy, whereas in Ni-Cu (in **b**), due to similar electronegativities, there is no correlation between E_v^f and Bader charge.

Fig. 2 | Model framework. The conventional framework to obtain E_v^f involves multiple DFT calculations starting with atomic relaxation of the supercell followed by E_v^f for every vacancy. This approach is computationally challenging the practicality of DFT calculations. The proposed framework eliminates DFT via machine learning. Structures are relaxed via CHGNet for DFT-level accurate lattice distortions via Model 1. Bader charges are then predicted for these structures via Model 2. The predicted Bader charges and relaxed structure from Models 1 and 2 are then fed into Model 3 to predict E_v^f s. DFT is bypassed and accurate E_v^f s are obtained with minimal computational cost.



electronegativities, it is added as a descriptor for each atom. The Wigner-Seitz volume is also added as a descriptor of the atom for which the Bader charge is predicted (see Methods). To ensure charge neutrality in these materials, we impose the physical constraint $\sum BC = 0$ by removing the mean predicted value from the original prediction in a given structure ($BC_{\text{constr},i} = BC_{\text{pred},i} - 1/N \sum BC_{\text{pred}}$).

We develop a modified CGCNN model optimized for atomic-level property prediction to predict Bader charges of atoms. Inspired by the CGCNN developed by Xie and Grossman¹⁹, which employs global pooling to aggregate atomic features into a single descriptor, we modify it by removing the pooling function, ensuring atomic representations remain distinct throughout the learning pipeline. This architectural change preserves local atomic interactions, improving the model's ability to learn structure-property relationships where atomic contributions vary spatially. These architectural modifications introduce minimal computational overhead while significantly improving the predictive resolution on the atomic scale. Eliminating the pooling operation increases the number of retained feature representations, but this is offset by optimizations in feature transformation and learning rate scheduling. The model maintains a parameter count comparable to that of the original CGCNN, ensuring scalability to large-scale material datasets without excessive computational demand.

By eliminating pooling operations, refining feature extraction mechanisms, and preserving atomic individuality, our model achieves a more granular and interpretable structure-property mapping. The selection of fixed-radius neighbors with Gaussian-expanded distance features (see Methods) ensures that structural subtleties are well represented, leading to improved predictive accuracy and generalization. These enhancements establish our approach as a robust framework for atomic-scale material property prediction, positioning it for applications where localized atomic interactions drive material behavior.

A comparison of Bader charge predictions with actual DFT values is shown in Fig. 4a. Here we can obtain accurate predictions following our PREDICT approach, training only on 70% of binary alloys (~1,360 Bader charges) and predicting in binary, ternary, and quaternary alloys (~3070 Bader charges), with $R^2 = 0.99$, RMSE = $1.59\text{e-}2$ e, and an absolute maximum error = $7.26\text{e-}2$ e. To compare the capability to predict the Bader charge in either a CHGNet or a DFT relaxed structure, we predict the Bader charges of the DFT structures and their equivalent CHGNet structures. Figure 4b, which gives $R^2 = 1.00$, RMSE = $5.32\text{e-}3$ e, and absolute maximum error = $3.60\text{e-}2$ e, shows that there is essentially no difference whether a CHGNet relaxed structure or a DFT relaxed structure is provided to the model - in either case, the predicted Bader charges are identical. This indicates that supercells obtained only from MLIP CHGNet can be uniquely used (and DFT can be bypassed) to obtain accurate Bader charges. Now, the last step is to predict the E_v^f s from CHGNet relaxed structures.

Vacancy formation energy predictions

Similarly to the Bader charge model, we use a modified version of CGCNN to predict E_v^f s. However, in contrast to the Bader charge model that uses the atom feature vector provided by CGCNN, we use a smaller atomic feature vector which only includes the atom type in the form of serialized group number and period number (see Methods). In comparison, the original atom feature vector consists of 92 components, whereas the one used for the E_v^f s consists of only 28 components, namely the group and period numbers of the periodic table to distinguish element type. This reduces complexity and allows for a better interpretation of the model when varying descriptors are included.

When training on 80% of the binary alloys and predicting in the quaternary alloys, Figure S6 shows that as volume or Bader charges are added to the atom feature vector, the prediction performance increases significantly. With the element type as the only initial atom feature vector, an R^2 of 0.18 is obtained as shown in Figure S6a. This improves from 0.18 to 0.36 with the inclusion of the volume of the vacancy atom, and then further improves to 0.74 with the inclusion of the Bader charges in Figures S6b and c, respectively. This indicates that at least for this type of model, including the Bader charge has the most significant influence. There is no further improvement by including the atomic volume in addition to Bader charge, which is expected because the bond lengths are already included in the model. Since the Wigner-Seitz volume is derived from the bond lengths, it becomes redundant information. Thus, for the subsequent models, we only use the shortened atom feature vector with the addition of vectorized Bader charges as input to the model.

The predictability of the model can be further improved by including the data of ternary alloys in the training set. By training on 80% of the binary and ternary data (~3110 E_v^f s) and testing on binary, ternary and quaternary data (~1300 E_v^f s), an $R^2 = 0.97$, RMSE = 0.05 eV, and absolute maximum error = 0.35 eV is achieved, as shown in Fig. 5a. Since good predictions are obtained using DFT-obtained Bader charges, we substitute the DFT-obtained Bader charges with the CGCNN predicted Bader charges from above. Figure 5b shows an R^2 of 0.96, RMSE of 0.05 eV, and absolute maximum error = 0.37 eV, indicating that there is minimal predictability loss when the predicted Bader charges are used instead of the DFT Bader charges.

As demonstrated in Fig. 4b, providing DFT or equivalent CHGNet relaxed structures to the Bader charge model results in essentially the same predictions indicating that CHGNet is a very good surrogate for DFT to obtain the relaxed structures. Figure 5b now further indicates that the use of predicted Bader charges as input to the E_v^f model also performs very well, and CGCNN surrogates DFT to obtain E_v^f s as well.

Now we bring all models together, i.e., the predicted structures from CHGNet and their predicted Bader charges are used as input to predict the E_v^f s from CGCNN. Thus, no DFT data is used as input to predict the E_v^f s. Figure 6a shows the comparison between DFT and the CGCNN (Bader

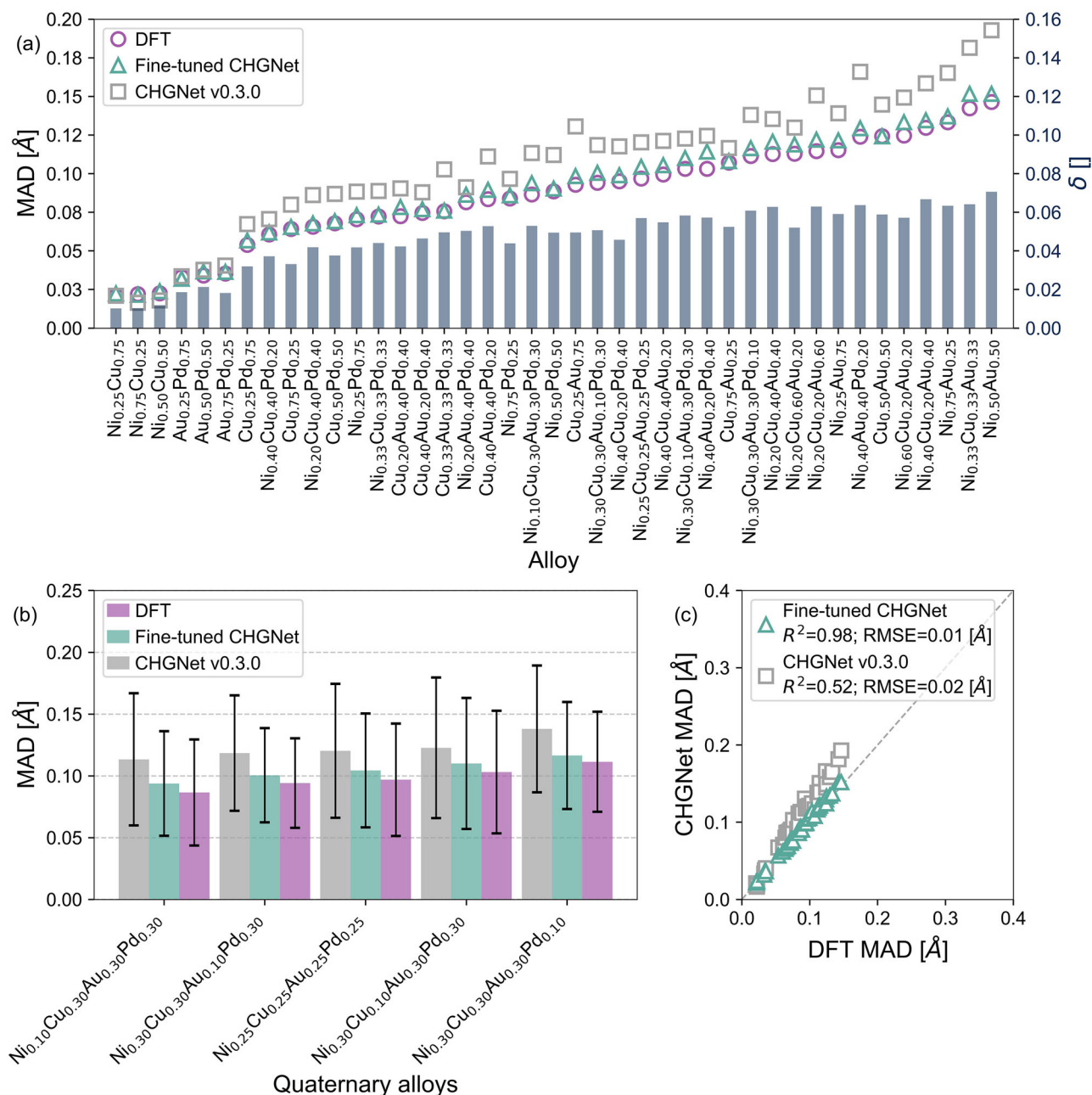


Fig. 3 | CHGNet lattice distortion. a Mean atomic displacement (MAD) (left axis) and δ (right axis) for all 41 binary, ternary and quaternary structures in this study. The MAD is shown with squares (CHGNet v0.3.0), triangles (fine-tuned CHGNet) and circles (DFT). The DFT and fine-tuned results are in very good agreement.

b MAD with standard deviation error bars for the five quaternary alloys. The fine-tuned CHGNet model shows much improved MAD and standard deviations.

c Comparison of MAD from DFT and CHGNet relaxations. The fine-tuned (triangles) results improve across all 41 alloys compared to CHGNet v0.3.0 (squares).

charge and E_v^f) models for CHGNet structures. An R^2 of 0.95, RMSE of 0.06 eV, and an absolute maximum error = 0.30 eV are achieved. Similarly to Fig. 5b, there is minimal predictability loss when starting from the CHGNet relaxed structures and predicting the E_v^f s. Figure 6b shows box plots for DFT and CHGNet + CGCNN BC + CGCNN E_v^f model results for the five Ni=Cu-Au-Pd alloys in this study. Thus, this indicates that DFT is efficiently bypassed to predict E_v^f s in FCC HEAs.

To further demonstrate the capability of these series of models, we take the equiatomic NiCuAuPd alloy and create the following randomized supercells: $4 \times 4 \times 4$ (256 atoms), $5 \times 5 \times 5$ (500 atoms), $8 \times 8 \times 8$ (2048 atoms), and $10 \times 10 \times 10$ (4000 atoms). The fine-tuned CHGNet model is used to relax each of these structures, as it is nearly impossible to do via DFT. However, due to limitations on computational resources, there are limitations as to how large a supercell may be created. Using an

NVIDIA 80GB A100 GPU, the largest supercell we could relax was the $10 \times 10 \times 10$ supercell containing 4000 atoms, which relaxed within around 7 min. This computational time is significantly shorter than the DFT simulations for the $3 \times 3 \times 3$ supercell containing only 108 atoms. Once CHGNet has relaxed the supercells, the Bader charges are then predicted and ultimately the vacancy formation energies are predicted for each atom using the respective CGCNN models. Figure 7 shows the average and standard deviations for each of these predicted vacancy formation energies in comparison to the $3 \times 3 \times 3$ (108 atom) supercell and calculations performed by DFT for this study. This indicates that these models can be used on system sizes much larger than is possible via DFT, with a high degree of accuracy. However, it should be noted that it remains difficult, if not impossible, to validate these results via conventional DFT results for such large supercells.

Fig. 4 | Bader charge predictions. **a** Bader charge predictions on all binary, ternary, and quaternary alloys by training on 70% of binary alloys data. Blue triangles are training data and purple circles are binary, ternary, and quaternary testing data. **b** Comparison of predicted Bader charges for both the DFT relaxed structures (x-axis) and fine-tuned CHGNet relaxed structures (y-axis). This indicates that either of the relaxed structures have essentially the same Bader charge predictions.

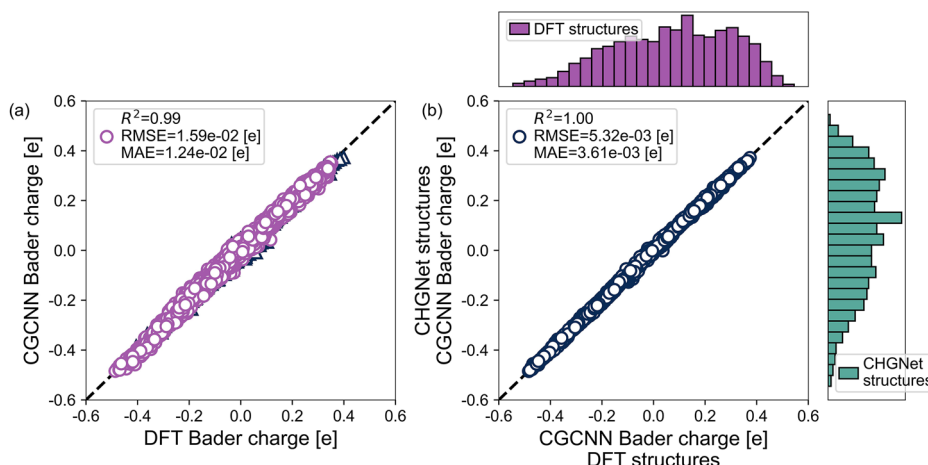


Fig. 5 | Vacancy formation energy predictions. **a** E_v^f predictions on all binary, ternary, and quaternary alloys by training only on 80% of binary and ternary alloys data. Blue triangles are training data and purple circles are binary, ternary, and quaternary testing data. **b** E_v^f predictions using the CGCNN predicted Bader charges as the input instead of DFT Bader charges of DFT relaxed structures.

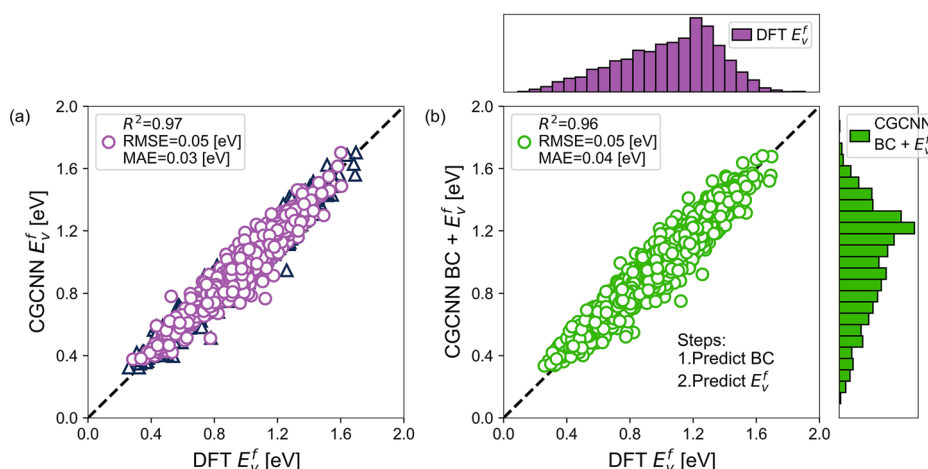
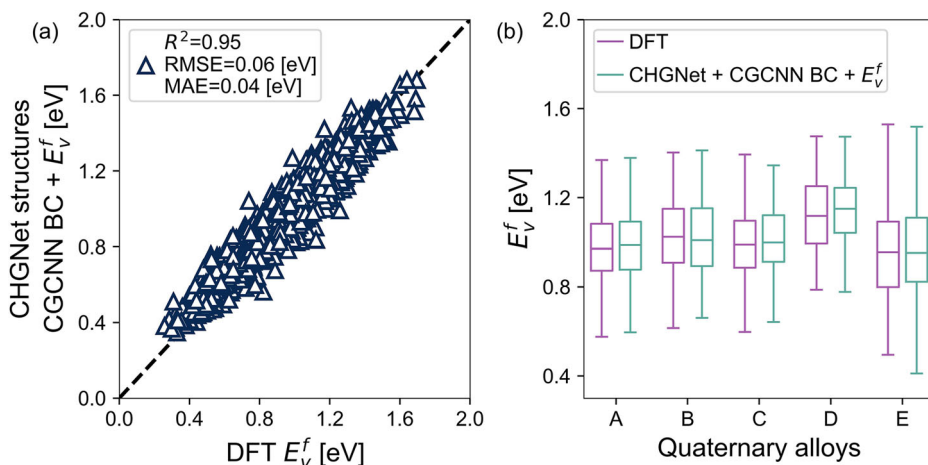


Fig. 6 | Bypassing DFT for vacancy formation energy predictions. **a** E_v^f predictions from the fine-tuned CHGNet relaxed structures by first predicting the Bader charges and using them as input to the E_v^f model in Fig. 5. **b** E_v^f box plots of the five Ni-Cu-Au-Pd alloys in this study comparing DFT results and CHGNet + CGCNN BC + CGCNN E_v^f .



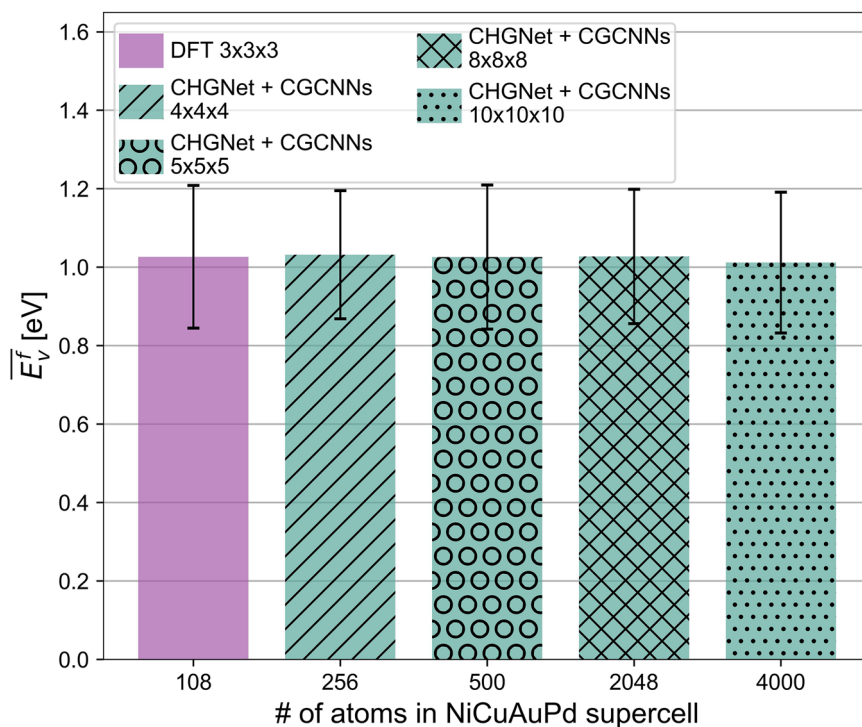
Model extension to unseen alloys

Broad applicability to unseen alloys is essential to establish the model's universality. However, it is well-known that ML models are inherently limited to their training environment. Here, we demonstrate that, with minor training in the unseen environment, a robust model is possible. Figure 8a shows that the current Bader charge model cannot predict Bader charges in random $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ 108 atom structures, shown as golden squares in Fig. 8. The predicted Bader charges for each atom/element in these alloys is around 0 e, even for the Ni

atoms. This is because, although Ni is present in the training set, its relative electronegativity is such that it always loses charge in the Ni, Cu, Au, and Pd system. In Ni-Co, Ni-Cr, and Ni-Co-Cr alloys, the electronegativities are as follows: $\chi_{\text{Ni}} > \chi_{\text{Co}} > \chi_{\text{Cr}}$. Consequently, Ni gains charge, and Co or Cr loses charge. In addition, because $\chi_{\text{Co}} > \chi_{\text{Cr}}$ in ternary alloys, Co gains charge when it is surrounded by more Cr atoms but loses charge when surrounded mainly by Ni. For the model to be accurate at predicting the Bader charges, instances where Ni and Co gain charge, and Co and Cr lose charge need to be included in the training dataset.

Fig. 7 | Model extension to larger supercells.

Average and standard deviation E_v^f results for equiatomic NiCuAuPd $3 \times 3 \times 3$ (108 atom) DFT calculations and the predictions for $4 \times 4 \times 4$ (256 atom, upward slanting hatching), $5 \times 5 \times 5$ (500 atom, open circle hatching), $8 \times 8 \times 8$ (2048 atom, x hatching) and $10 \times 10 \times 10$ (4000 atom, dot hatching) supercells via the fine-tuned CHGNet, CGCNN Bader charge, and the CGCNN E_v^f models.



To remedy the poor extrapolative properties of the Bader charge model, we include the computed Bader charge values from the $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ structures as additional data for fine tuning. The Bader charge data from the 50-50 constituent binary alloys of the Ni-Cu-Au-Pd system is also included to mitigate knowledge loss. During fine tuning, freezing the pretrained weights of model layers and decreasing the learning rate also help mitigate this loss. We choose to freeze the middle layers and only allow the weights of the final output layer and the `node_embedding`, `edge_embedding`, and `atom_nbr_embedding` layers to update. This is because structures containing unseen elements and bond types are introduced into the dataset. After fine-tuning on 70% of this reduced dataset (i.e., only 50-50 binaries plus the Ni-Co, Ni-Cr, and Ni-Co-Cr random alloys resulting in ~680 training datapoints, with only 3 new simulations for the Ni-Co-Cr system) for only 50 epochs with a batch size of 25 and a reduced learning rate of 0.005, an $R^2 = 0.99$ and $\text{RMSE} = 2.32\text{e-}2$ e is achieved, as shown in Fig. 8b. Thus, with minimal simulations (i.e., one simulation per extra structure for a total of three simulations) and minimal training steps, the models improve significantly, allowing for predictions in the Ni-Co-Cr alloy system. Furthermore, there is negligible knowledge loss against the previous Ni-Cu-Au-Pd data, shown as purple circles in Fig. 8, as R^2 and RMSE have minimal changes from Fig. 4a.

Similar updates are needed in the E_v^f models as shown in Fig. 8c. This is because the model is trained on E_v^f s ranging from ~0.4 to 1.5 eV, whereas those in the Ni-Co-Cr system range from ~1.5 to 2.2 eV.

The model is updated similar to the Bader charge model. We unfreeze the last convolutional layer in addition to the layers mentioned above. Fine tuning is performed using the E_v^f s of the $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ structures in the training set, as well as the binary and ternary Ni-Cu-Au-Pd constituent alloys. Another way to maintain prior knowledge is to maintain the learning rate and reduce the batch size. We maintain the learning rate of 0.001 for fine tuning the E_v^f model. However, the batch size is reduced from 128 to 28. Figure 8d shows results for fine tuning over 300 epochs on 75% of the new dataset, resulting in $R^2 = 0.97$ and $\text{RMSE} = 0.06$ eV for all predictions. Thus, this fine tuning drastically improves the predictability in these alloys and maintains the predictability in the previous alloys with minimal additional DFT calculations. These two

fine-tuned models can now be confidently applied to other alloys within the Ni-Co-Cr system.

We further test the model against the Ni-Co-Cr alloys from MPContribs provided by Walsh et al.²⁶, shown as blue triangles in Fig. 8. These are FCC alloys containing 108 atoms that consist of quasirandom structures and structures with varying degrees of short-range ordering (SRO). First, to validate the performance of our fine-tuned Bader charge model, we calculate the Bader charge for ten different structures from this dataset. These structures are all equiatomic, with varying degrees of SRO. Figure 8a shows that the initial Bader charge predictions for these Ni-Co-Cr alloys (triangles) have predictions similar to the previous $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ alloys (squares). With the fine-tuned Bader charge model, significant improvements occur, as shown in Fig. 8b.

Second, to verify the E_v^f model, the E_v^f s are calculated in a structural 0.5 SRO structure and a quasirandom structure from the above MPContribs dataset (i.e., 218 DFT calculations for E_v^f s). Figure 8c shows the predictions of the E_v^f s with the initial model. Similarly to the $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ alloys (squares) in Fig. 8c, the values for these quasirandom and SRO alloys (triangles) are underestimated. Using the fine-tuned model, Fig. 8d shows that the E_v^f s are predicted nearly identically to the $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ alloys. Thus, these CGCNN models can be used to predict Bader charges and E_v^f s within a Ni-Co-Cr alloy system for both quasirandom and SRO structures.

Discussion

In this study, we develop an ML framework to eliminate the need for DFT calculations to predict E_v^f s in HEAs. By fine-tuning CHGNet with simple 16-atom structures, we show that DFT-level accuracy in lattice distortions can be predicted in larger, 108-atom, binary, ternary, and quaternary structures. The CHGNet relaxations for the perfect structures are achieved within a few seconds with less computational resources than the VASP equivalent, which requires several hours to days to complete. Furthermore, we predict the partial charges in the form of Bader charge for each atom with a high degree of accuracy. The predicted Bader charges are then fed into another model that predicts the E_v^f s.

Due to the limited availability of E_v^f data followed by computationally intensive calculations to generate such a database, we generate a database of

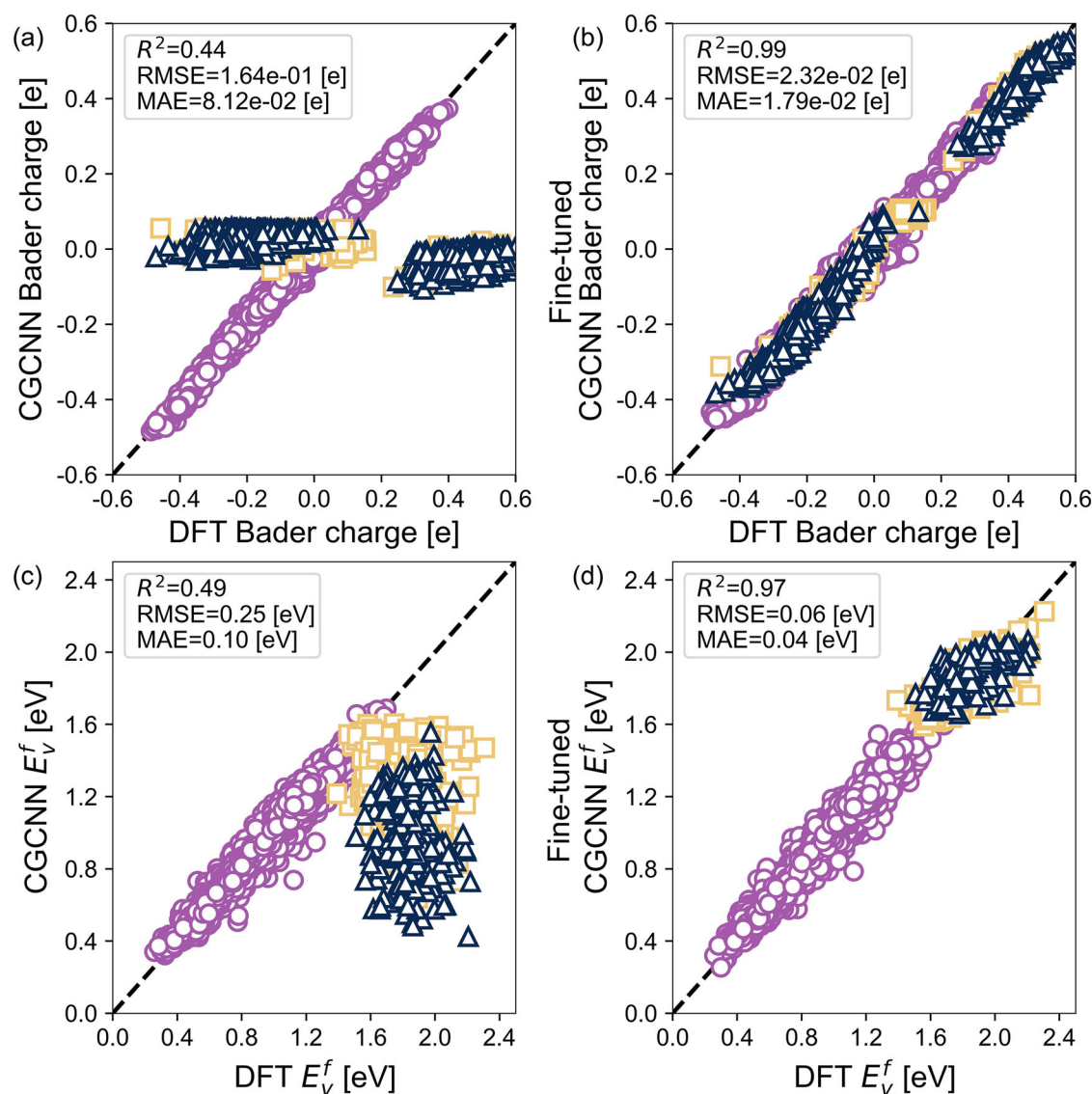


Fig. 8 | Model fine tuning and validation in unseen Ni-Co-Cr alloys. The data in Ni-Cu-Au-Pd system is in purple circles, in $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ random alloys is in gold squares, and the quasirandom and varying short-range ordered (SRO) structures in $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ alloys in blue triangles is from the structure database deposited in MPContribs by Walsh et al.²⁶. **a** Bader charge predictions on all alloys using the model trained on binary Ni, Cu, Au, and Pd data. **b** Bader charge predictions on all alloys with the fine-tuned Bader

charge model. **c** E_v^f predictions on all alloys using the model trained on binary and ternary Ni, Cu, Au, and Pd alloys. **d** E_v^f predictions on all alloys with the fine-tuned E_v^f model. Fine tuning is achieved by training on binary and ternary Ni, Cu, Au, and Pd alloys and $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$ random alloys from this study. By fine-tuning with Ni-Co-Cr data, E_v^f s are accurately predicted in **(d)** compared to **(c)**.

only 41 alloys in a Ni-Cu-Au-Pd FCC alloys system. These consist of binary, ternary, and quaternary alloys with 108 atoms each. The Bader charge is also calculated for each structure, resulting in a total Bader charge and E_v^f database of 4428 datapoints each. Choudary et al.²⁰ used a database of only 530 datapoints to validate their use of the ALIGNN MLIP. They used the pre-trained ALIGNN model to relax the vacancy structure to obtain the E_v^f s. However, they found that the model could not handle defect relaxations as they were not present in the training set. We attempt a similar approach using a CHGNet model, which is fine-tuned with vacancy structures. Even with this fine-tuning, we find that the vacancy would occasionally migrate to neighboring atom sites. This results in poor E_v^f predictions as the NN environments change before and after creating a vacancy. Thus, until these or other MLIPs are trained on a database containing defects, with roughly the size of the databases on which the originals were trained ($\sim 146,000$ structures), caution should be exercised when analyzing the energetics of a defect via direct energy predictions. We overcome the

deficiency of MLIPs to relax structures with defects by using the MLIP to relax a defect-free structure. This defect-free structure is then fed into simpler, modified CGCNN models to accurately predict the Bader charges and ultimately the E_v^f s for each atom. Thus, DFT can be bypassed.

Witman et al.³⁰ created a database consisting of ~ 1500 defect sites from ~ 200 oxide structures. They trained their defect graph neural network (dGNN) with a defect-wise three-fold cross-validation that resulted in their best model performances. They were able to obtain a mean absolute error (MAE) of 0.30 ± 0.03 eV and 0.65 ± 0.08 eV for the E_v^f s of oxygen and other elements, respectively. In this paper, by training on ~ 3100 E_v^f s from 36 total binary and ternary alloys and predicting on all binary, ternary and quaternary alloys results in the MAE = 0.03 eV, which is an order of magnitude improvement. Even when training on ~ 1500 E_v^f s from 18 binary alloys and predicting on quaternary alloys results in the MAE = 0.08 eV. A major difference between dGNN and the E_v^f model presented in this paper is that we include Bader charges, eliminate pooling operations, refine feature

extraction mechanisms, and preserve atomic individuality. This allows for improved predictive accuracy and a model that is interpretable.

Although these models do well in predicting Bader charges and E_v^f s in unseen binary, ternary, and quaternary alloys within the same system, the extrapolative properties may not be fully explored. Jain suggests that there are no clear standards within the ML community to quantify such extrapolation³¹. In an attempt to demonstrate the extrapolation of the models, we predict in unseen $\text{Ni}_{0.50}\text{Co}_{0.50}$, $\text{Ni}_{0.75}\text{Cr}_{0.25}$, and $\text{Ni}_{0.33}\text{Co}_{0.33}\text{Cr}_{0.33}$. We show that a small amount of fine-tuning can result in large improvements in predictability while the models maintain previous knowledge. Furthermore, we demonstrate that these fine-tuned models can accurately predict Bader charges and E_v^f s in quasirandom and SRO Ni-Co-Cr alloys from such a database deposited in MPContribs by Walsh et al.²⁶. This adaptation to new systems exploits the similar FCC structure of both systems, as all models present are structural- (and physics) based models (i.e., graph neural networks). Future studies could be undertaken to determine the amount of new data needed for the best predictions without too much computational expenses. In addition, studies that remove alloys with certain elements (e.g., Ni) from the main dataset presented here and predict on them could also be done to observe the effects of similar Bader charge and E_v^f ranges on the predictions.

Compared to other models, not relying on DFT calculations is another major advancement. For example, Witman et al.³⁰ and Cheng et al.²¹ reduced the need for expensive vacancy DFT calculations by predicting the E_v^f s on the perfect DFT relaxed structure. However, as mentioned earlier, this still remains time consuming and computationally expensive, especially for complex materials such as HEAs. By drastically reducing the computational load via the fine-tuned CHGNet, one could now confidently and quickly relax structures within the large HEA phase space, also including larger supercell sizes as demonstrated in Fig. 7. The defect energetics of this phase space can also be confidently predicted via our structure- and charge-informed E_v^f model. This allows for a faster, broader, “high-throughput” approach to be applied to identify materials with desired defect properties where a similar DFT-based search would require a significant amount of computational time and resources.

Albeit this study focuses on defect energetics, the approach to fine-tuning CHGNet to obtain atomic displacements is an achievement in itself, as the model can be used in other relevant studies that rely on structural information. This includes correlation between the normalized yield strength and mean squared atomic displacement (MSAD) by Okamoto et al.¹¹, where the authors directly correlated LLD with strength in CrMnFeCoNi HEA and some of its constituent ternary and quaternary alloys. Thus, mechanical properties can be obtained quickly from the atomic displacements captured by a fine-tuned CHGNet model.

Methods

First-principles calculations and database

DFT calculations are performed using the Vienna ab initio simulation package (VASP)^{32–34}. The projector-augmented wave (PAW) method is employed with the exchange–correlation energy evaluated by generalized-gradient approximation (GGA) using the Perdew–Burke–Ernzerhof (PBE) functional³⁵.

Structure relaxation dataset for CHGNet fine-tuning

$2 \times 2 \times 1$ FCC supercell, 16-atom, randomized DFT relaxed structures are taken from our previous works^{15,29}, where a plane-wave cutoff of 500 eV and a Monkhorst–Pack k-point grid of $6 \times 6 \times 12$ were used for all calculations. 10^{-8} eV was chosen as the electronic self-consistency energy convergence criterion. -10^{-4} eV/Å was chosen as the force convergence criterion of the ionic relaxation. This dataset consists of binary, ternary, quaternary, and quinary alloys within the Ni, Cu, Au, Pd, and Pt alloy system. However, only binary and ternary alloys consisting of ~ 1400 structures from this dataset are used to fine-tune the CHGNet v0.3.0 model, as described later.

Bader charge and vacancy formation energy dataset

Bader charges and E_v^f s are calculated in randomized binary, ternary, and quaternary $3 \times 3 \times 3$ FCC supercells containing 108 atoms in the Ni-Cu-Au-Pd alloy system. A total of 41 supercells are created, consisting of 18 binary supercells, 18 ternary supercells, and 5 quaternary supercells. We follow the DFT calculations explained in our previous work²⁷, where a plane-wave cutoff of 400 eV and a Monkhorst–Pack k-point grid of $4 \times 4 \times 4$ are used. 10^{-4} eV is chosen for the electronic self-consistency energy convergence criterion, and -10^{-2} eV/Å is chosen as the force convergence criterion of the ionic relaxation. The partial atomic charges (i.e., Bader charges) of all atoms in defect-free supercells are obtained by using Bader analysis^{36,37} performed on the charge-density files in VASP. This results in a dataset of 4428 Bader charges.

After the atom positions and the supercell size and shape are allowed to relax in the defect-free supercell, the vacancy simulations are completed by removing each atom individually and relaxing the defect supercell with a fixed supercell size and shape while relaxing the atom positions. The E_v^f s of each atom are calculated using Eq. (1) as follows:

$$E_v^f = E_D - E_0 + \mu_X \quad (1)$$

where E_v^f is the vacancy formation energy, E_D is the energy of the supercell containing the vacancy, E_0 is the energy of the perfect supercell, and μ_X is the chemical potential of element X. Following the approach by Manzoor et al.¹⁴ and Zhang et al.³⁸, μ_X is calculated by running all 108 vacancy simulations for each supercell to obtain the average E_v^f , shown in Eq. (2)

$$\overline{E_v^f} = \frac{1}{N} \sum_{k=1}^N E_D^k - \frac{N-1}{N} E_0 \quad (2)$$

where $\overline{E_v^f}$ is the average vacancy formation energy, E_D^k is the energy of the defect supercell for each vacancy site k , and N is the total number of atoms in the supercell. μ_X is calculated for each element via E_v^f using Eq. (3) as follows:

$$\mu_X = E_0 + \overline{E_v^f} - \frac{1}{N_X} \sum_{j=1}^{N_X} E_j^X \quad (3)$$

where E_j^X is the energy of the defect supercell with the element X removed for each vacancy site j , and N_X is the number of X element atoms present in the supercell. Calculating the E_v^f for each atom in all supercells results in a total of 4428 datapoints. References¹⁴ and³⁸ contain the validation and details of this approach.

CHGNet fine-tuning

Model fine-tuning is needed to adapt the CHGNet v0.3.0 model to predict relaxed structures with the correct lattice distortions. CHGNet v0.3.0 was trained on Materials Project data consisting of $\sim 146,000$ inorganic materials^{22,25}. These materials were simulated by DFT with VASP using parameters different from those used in this study. The energies given by VASP in this study and those predicted by CHGNet v0.3.0 for the same DFT structures are different. Thus, the energies must be corrected for fine-tuning. We use CHGNet to predict the energies of the pure elements. Then, using a linear relationship with the energy differences of the pure elements, the energies of the alloys are corrected. This is achieved using Eq. (4) as follows:

$$E_{\text{corr}} = E_{\text{DFT}} + \sum C_X \Delta E_X \quad (4)$$

E_{corr} is the corrected energy/atom used to fine-tune CHGNet, E_{DFT} is the energy/atom of our VASP simulation for the structure, C_X is the composition in atomic percentage of the element X, and ΔE_X is the difference in energy/atom of element X in pure metal using the VASP parameters mentioned above for this study and that given by CHGNet v0.3.0. Mathematically, this is given as $\Delta E_X = E_{\text{v0.3.0,X}} - E_{\text{DFT,X}}$. A

representation of this for Ni-Cu and Ni-Au alloys from the 16-atom structures is given in the Supplementary Fig. S1a–d. Although a correction could be done for each structure using ΔE_{alloy} , this is computationally expensive as each structure must be loaded and predicted on with CHGNet. Thus, we choose to use the simpler and less computational approach of the linear relationship in Eq. (4). These ΔE_X are also saved to remove this correction to compare the predicted energy/atom from the fine-tuned model and our DFT calculations. However, this is not within the scope of this study, and we do not report such values here.

The dataset used to fine-tune CHGNet consists of the 16-atom binary and ternary alloys (~1400 structures) and only 20% of each of the binary defect structures (~390 structures). Thus, the fine-tuning dataset is roughly 1% of the size of the data on which CHGNet v0.3.0 was trained. The CHGNet v0.3.0 model is loaded and we fine-tune it on 75% of our small dataset, with 25% being held for validation. We set targets as energy/atom and forces/atom (eF) to only minimize the losses. Because we update the energies via Eq. (4) to match what CHGNet v0.3.0 already predicts, the model minimizes the forces/atom present within the materials. This allows the model to better capture the lattice distortions present. The weights of the following layers of the model are frozen (i.e., do not update with training) per recommendations made on GitHub at <https://github.com/CederGroupHub/chgnet>: atom_embedding, bond_embedding, angle_embedding, bond_basis_expansion, angle_basis_expansion, atom_conv_layers[: -1], bond_conv_layers, and angle_layers. The optimizer is Adam, with the criterion as mean squared error (MSE), $MSE = 1/N \sum_{i=1}^N (y_i - \hat{y}_i)^2$. Here, y_i represents the ground truth atomic-level property, and $\hat{y}_i$ is the predicted value of the network. The learning rate is 0.01. Fine-tuning is performed for 50 epochs, with the train/validation loss plots shown in Supplementary Fig. S1e.

The mean atomic displacement (MAD) is used to validate that the fine-tuned model captures the lattice distortions present within these random alloys. This is quantified by calculating the displacement of each atom from its corresponding, undistorted ideal lattice site. For FCC, the unit cell lattice sites in fractional coordinates are (0, 0, 0), (0.5, 0.5, 0), (0.5, 0, 0.5), and (0, 0.5, 0.5). The MAD is calculated as follows:

$$MAD = \frac{1}{N} \sum_{j=1}^N \left(\sqrt{(x_i - x_d)^2 + (y_i - y_d)^2 + (z_i - z_d)^2} \right)_j \quad (5)$$

where i and d are the ideal lattice site and distorted coordinates, respectively. N is the number of atoms in the structure, and this is summed over all atoms j . As the difference in radii from the elements present in the alloy grows, so does the MAD. To quantify this for an arbitrary number of elements, we use δ , which is an average difference in atomic radii in multicomponent alloys, quantified by Zhang et al.²⁸ It is calculated as

$$\delta = \sqrt{\sum_{i=1}^N c_i (1 - r_i/\bar{r})^2} \quad (6)$$

where N is the number of elements in the alloy, c_i is the atomic percentage of element i , $\bar{r} (= \sum c_i r_i)$ is the average atomic radius, and r_i is the atomic radius taken from ref. 39.

Crystal graph convolutional neural network (CGCNN) graph construction and feature representation

To refine feature extraction, we replace Softplus activations with ReLU, improving gradient flow and convergence stability. Additionally, LeakyReLU is integrated into the neighbor interaction module to enhance feature discrimination, particularly in low-energy atomic environments where feature suppression could lead to information loss. To further optimize training dynamics, we introduce a CyclicLR scheduler, dynamically adjusting the learning rate to avoid premature convergence and stagnation. These modifications collectively improve the expressiveness of the model,

enabling it to capture atomic-scale variations more effectively than the original CGCNN by Xie and Grossman¹⁹.

Each crystal is represented as an undirected graph, where the atoms serve as nodes, and the edges correspond to interatomic interactions. In this work, rather than using the entire structure as the graph, a “mini” graph is used. That is, a graph consisting of only the central atom (the atom for which predictions are made) and its 12 NNs in FCC. The feature representation is structured as follows: 1) Each atom is assigned a feature vector encoding its chemical and structural properties. 2) Each bond is represented by a neighbor feature vector, capturing interatomic distances and local bonding characteristics. Given an atomic feature matrix $h \in R \wedge (N \times F)$, where N is the number of atoms and F is the feature dimension, the network iteratively updates the atomic states via convolutional transformations:

$$h_i^{(l+1)} = \sigma \left(W \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} | j \in N(i)\}, e_{ij} \right) \right) \quad (7)$$

where AGG is an aggregation function over neighboring atomic states, W is a trainable weight matrix, and σ is the ReLU activation function.

Unlike standard CGCNNs¹⁹, where atomic features are globally pooled:

$$h_{\text{crystal}} = \frac{1}{N} \sum_{i=1}^N h_i^{(L)} \quad (8)$$

we retain per-atom embeddings, allowing predictions at the atomic level rather than a bulk-crystal approximation. This shift enhances interpretability and improves predictive accuracy for properties governed by localized atomic interactions rather than global material characteristics.

CGCNN neighbor selection and local environment encoding

The neighborhood of each atom is determined using a fixed-radius approach, where all atoms within a cutoff distance, r_{cut} , are included as neighbors, up to a maximum of 12. Unlike adaptive methods that adjust on the basis of bond strengths or chemical significance, our approach maintains computational efficiency and consistency across materials with varying coordination environments.

To ensure smooth feature transformations, Gaussian basis expansion is applied to interatomic distances:

$$e_{ij} = \exp \left(-\frac{(d_{ij} - \mu_k)^2}{(2\sigma^2)} \right), \text{ for } k = 1, \dots, K \quad (9)$$

where d_{ij} is the Euclidean distance between atoms i and j , and μ_k represents Gaussian centers distributed over a predefined distance range. This transformation allows the model to learn continuous relationships between atomic separations and material properties, improving sensitivity to subtle structural variations.

CGCNN data preprocessing and normalization

To maintain numerical stability and prevent feature scale disparities, atomic features are min-max normalized to the [0,1] range:

$$h_i = \frac{h_i - \min(h)}{\max(h) - \min(h)} \quad (10)$$

Bond feature vectors undergo standardization to ensure uniform representation across different material classes. Once normalized, the values are then one-hot encoded to a vector with a user-determined number of bins. These preprocessing steps align with our architectural goal of retaining fine-grained atomic detail throughout the learning pipeline. Unlike traditional CGCNN-based models, which merge atomic representations into bulk-crystal descriptors, our method maintains individual atomic embeddings, enhancing interpretability and predictive accuracy.

CGCNN training Bader charge and vacancy formation energy CGCNNs

The models are trained using the MSE loss function. Optimization is performed using the Adam optimizer. To train the Bader charge model, an initial learning rate of 0.01 is set and dynamically controlled using the CyclicLR scheduler. This model is trained only on the 108 atom binary structures, with a total dataset size of 1944. We partition this dataset into 70% training, 15% validation, and 15% test splits to ensure generalizability across unseen materials. The model is trained for 400 epochs with a batch size of 200. We increase the convolutional depth to three layers, allowing for deeper hierarchical feature extraction, capturing both short- and long-range interactions. For Bader charge, the atom feature matrix h_i consists of the original atom feature matrix of Xie and Grossman¹⁹ that consists of information such as element type, weight, volume, etc. In addition to this information, we append the min-max normalized volumes and electronegativities using Eq. (10) to the atom feature matrix. Here, 10 and 20 Å³ are the minimum and maximum used for the volume, respectively. 0.7 and 4 are used as the minimum and maximum for electronegativities, respectively. Both are one-hot encoded into 64 bins. This model architecture results in ~44,500 trainable parameters. These are then passed to the model architecture for training and validation, along with the structure.

The model setup for the E_v^f is similar to that for the Bader charge. An initial learning rate of 0.001 is set and dynamically controlled using the CyclicLR scheduler. This model is trained on the 108 atom binary and ternary structures, with a total of 3888 datapoints. We partition this dataset into 80% training, 10% validation, and 10% test splits to ensure generalizability across unseen materials. The model is trained for 400 epochs with a batch size of 128. The convolutional depth is set to four layers. However, the atom feature matrix is now only the first 28 components of the original atom feature matrix of Xie and Grossman¹⁹. This consists of only the period and group numbers of the elements to determine the element type. This model architecture results in ~45,800 trainable parameters. In addition to this modified atom feature matrix, min-max normalized Bader charges are appended. Here, -0.5 and 0.5 are used as minimum and maximum, respectively. These are then one-hot encoded into only 32 bins, following the analysis provided in Supplementary Fig. S6. The updated atom feature matrix, along with the structure, are then passed to the model for training and validation.

CGCNN fine-tuning Bader charge and vacancy formation energy models

To test the generalization and predictability of the models, we use VASP to generate Bader charge and E_v^f values for three random structures containing 108 atoms. These are Ni_{0.50}Co_{0.50}, Ni_{0.75}Cr_{0.25}, and Ni_{0.33}Co_{0.33}Cr_{0.33}. These are FCC structures, and we follow the VASP calculation parameters explained in the Bader charge and E_v^f dataset section above.

For further validation of the Bader charge predictions, we calculate Bader charges for 10 structures ranging from quasirandom to SROs from a short-range order structure database deposited in MPContribs (https://portal.mpcontribs.org/projects/mpca_sro/) by Walsh et al.²⁶ We then take one SRO, a structural 0.5, and one quasirandom structure and calculate the E_v^f s for all 108 atoms. These values are then used to validate the E_v^f predictions of the fine-tuned model.

Data availability

The data, codes, and models for reproducing these results can be found here: https://github.com/Materials-Computation-Data-Science-MCDC/CGCNN_VFE.

Code availability

The underlying code and training/validation datasets for this study are available in the CGCNN_VFE repository and can be accessed via this link https://github.com/Materials-Computation-Data-Science-MCDC/CGCNN_VFE.

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Author contributions

N.L., D.S.A., and P.S. conceptualized the framework. D.S.A. acquired funding. N.L. performed DFT calculations. N.L. and P.S. developed the codes for this work and wrote the main manuscript text. N.L. and D.S.A. edited the manuscript. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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